Neural Computing and Applications (2020) 32:13479–13490
https://doi.org/10.1007/s00521-020-04755-4

ORIGINAL ARTICLE

Efficient and hardware-friendly methods to implement competitive learning for spiking neural networks

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Received: 8 August 2019 / Accepted: 23 January 2020 / Published online: 3 February 2020
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Abstract

Spiking neural network (SNN) trained by spike-timing-dependent plasticity (STDP) is a promising computing paradigm for energy-efficient artificial intelligence systems. During the learning procedure of SNN trained by STDP, another two bio-inspired mechanisms of lateral inhibition and homeostasis are usually implemented to achieve competitive learning. However, the previous methods to implement lateral inhibition and homeostasis are not designed with hardware in mind, resulting in solutions that are not efficient for deployment on neuromorphic hardware. For example, the existing lateral inhibition methods induce a great number of connections that are proportional to the square of the number of learning neurons. The classical homeostasis methods depend on the fine-tuned membrane threshold with no hardware solution provided. In this paper, we propose two hardware-friendly and scalable methods to achieve lateral inhibition and homeostasis. Using only one inhibitory neuron for one learning layer, our proposed lateral inhibition method can reduce inhibitory connection number from N^2 to $2N$ and hardware overhead by sharing refractory control circuits. Utilizing the adaptive resistance of memristor, we propose a novel homeostasis method through adapting the leaky current of spiking neurons. In addition, the learning efficiency of different homeostasis methods are studied for the first time by simulating on the cognitive task of digital recognition of MNIST dataset. Simulation results show that our proposed homeostasis method can improve the learning efficiency by 30–50% while maintaining the state-of-the-art performance.

Keywords Spiking neural network · Homeostasis · Lateral inhibition · Competitive learning · Memristor

1 Introduction

Spiking neural networks (SNNs) are the third-generation artificial neural networks (ANNs) that aim to achieve artificial intelligence by mimicking the computation and learning methods of brain [1, 2]. Compared with the second generation of ANNs, the deep neural networks (DNNs), SNNs are more powerful in terms of energy efficiency and robustness [3–5]. Previous studies have shown that SNNs can be trained by spike-timing-dependent plasticity (STDP) to extract features in an unsupervised manner [6, 7]. Only based on the spike timing relation between two connected neurons, STDP can be computed in a local event-driven manner, which is intrinsic energy-efficient when implemented on hardware [8, 9]. SNNs trained by STDP provide efficient solutions for artificial intelligence applications, especially for power-critical ones including mobile platforms, IoT, and drones [10–12]. For simplicity, the learning algorithm of SNN trained by STDP is referred as SNN-STDP in this paper.

Though with many intrinsic advantages, the applications of SNN-STDP have been long challenged due to the low performances on machine-learning tasks [13]. Querlioz et al. proposed the first unsupervised SNN-STDP learning algorithm that achieved comparable performances on the task of classification of MNIST dataset [14]. In addition to

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STDP, the methods of lateral inhibition and homeostasis were merged to realize the competitive learning mechanism [15, 16]. STDP rules the synapse weight update quantity, while lateral inhibition and homeostasis work together to force neurons learn different patterns with the same learning progress. Lateral inhibition can force different neurons to learn different patterns by inhibiting other neurons from firing when one neuron has already responded to the input. Without lateral inhibition, learning neurons in fully connected architecture or neuronal maps in convolutional architecture in the same layer will learn the same pattern due to the same connection to pre-layer. After presenting several training examples, some learning neurons will fire more frequently due to the inhomogeneity of input patterns, while the others are always inhibited from learning due to the lateral inhibition. Indeed, lateral inhibition exacerbated the inhomogeneity. Therefore, homeostasis method is added to reduce the inhomogeneity and make sure that the neurons in the same layer have the same opportunity to learn when presenting new inputs [17, 18]. The existing homeostasis method is based on membrane threshold adaptation, which increase the firing threshold of high-firing-rate neurons. Thus, every learning neurons in the same layer will fire and learn at nearly the same rate under the control of homeostasis method. After the unsupervised learning procedure, every neuron in the learning layer learns a pattern that is more sensitive to one specific pattern, and one pattern can be learned by many neurons in one layer because of the pattern variation. To perform the inference, neurons in the learning layer will be labeled as one class according to their highest firing rate for the specific class using a training subset. Then the labeled neurons can be used for inference through voting for class they labeled.

Currently, lateral inhibition and homeostasis are adopted in most of the unsupervised learning algorithms of SNN-STDP with feed-forward and convolutional network architectures [19–24]. However, the hardware overhead and feasibility of the existing method to achieve lateral inhibition and homeostasis are not considered carefully when explored in software simulation. There are two methods to apply lateral inhibition. The first method required every learning neuron (excitatory neuron) to connect to each other in the same layer for transmitting inhibition spikes when one has responded to the input [15]. The second method added the same number of inhibitory neurons as the excitatory neurons. Every excitatory neuron is connected to its corresponding inhibitory neuron, and every inhibitory neuron is connected to all the excitatory neurons except the one from which it receives a connection [19]. Both the two lateral inhibition methods necessitated a great number of connections, which is proportional to the square of the number of the excitatory neurons. In this paper, we simplify the way to achieve lateral inhibition by using only one inhibitory neuron for one learning layer. If one learning neuron is excited, it will send a strong spike to excite the inhibitory neuron, which will send inhibitory spikes to all excitatory neurons in the same layer and force them to run into refractory period. Using our proposed method, only one suit of circuits is needed for refractory period control, and the connection number for lateral inhibition is reduced from N^2 to $2N$.

The existing homeostasis methods are based on membrane threshold adaptation (MTA), which has been shown to perform well in software simulation. However, MTA necessitates a fine-tuned voltage threshold for every learning neuron which is costly for hardware implementation. Until now, no hardware solutions have been provided for MTA-based homeostasis methods [15, 19, 24, 25].

Recently, memristor has been widely explored for the hardware implementation of artificial synapses for low-power neuromorphic systems [26, 27] With gradually adaptive conductance with nonvolatility, only one memristor device is needed to implement STDP learning, leading to significant power and area reduction compared with solutions using transistors [28–30]. To further explore the intrinsic advantages of memristor, we propose a novel method to achieve homeostasis through adapting the leaky current. The proposed method adds an special artificial neuron to connected to learning neurons in one layer via memristors, and the conductance of the memristor can be programmed by spike voltage to adapt the leaky current of every learning neurons. Only one memristor for per learning neuron and one extra artificial neuron is needed to achieve homeostasis in one learning layer. Since the complexity of hardware implementation of MTA is unknown, our proposed method may be a promising solution for homeostasis implementation.

We deploy our proposed methods in a network of fully connected single-layer architecture for performance evaluation. Software simulations are carried out in Brian simulator on the task of digital recognition of MNIST dataset. The rest of this paper is organized as follows. In Sect. 2, we propose our methods to achieve lateral inhibition and homeostasis and discuss the hardware-implementation advantages of our proposed methods. In Sect. 3, the neural models of SNN-STDP for simulation are introduced. Section 4 presents simulation results and studies on the learning efficiency of different homeostasis methods. Finally, conclusions are drawn in Sect. 5.

2 The proposed competitive learning methods for SNN-STDP

2.1 One-neuron lateral inhibition method

The number of connections for lateral inhibition using the previous methods is proportional to the square of the number of excitatory neurons in one layer. The reason for the massive connections is that previous lateral inhibition method demands connections from N neurons to N (or $N-1$) neurons. To reduce the connection number, we propose a lateral inhibition method thats use only one inhibition neuron for implementing lateral inhibition in one layer. The schematics of the proposed lateral inhibition method is plotted in Fig. 1.

All excitatory neurons in one layer are bi-directionally connected to the inhibitory neuron. Therefore, the connections for lateral inhibition are from N neurons to one neuron, and the connection number is $2N$ that is far less than N^2. When the excitatory neuron is activated by the input pattern, it sends spike to the inhibition neuron except for spikes to the next layer. Only one spike is enough to activate the inhibitory neuron immediately, then the inhibitory neuron sends inhibition spikes to all the connected excitatory neurons. When the excitatory neuron receives the inhibition signal, its membrane potential will be reset to the resting level and turn into the refractory period. That means that all the excitatory neurons in the same layer will enter the refractory period at the same time after one of them responds to the input pattern. After the refractory period, all the excitatory neurons will return to the normal state simultaneously. The refractory period is set to be 10 ms in our simulation.

Because the inhibition operation is implemented based on the refractory control circuits of the excitatory neuron, the proposed inhibition method does not introduce extra hardware overhead. What is more, there only needs one circuit of the refractory period control for all the excitatory neurons in the same layer. All in all, the proposed method can reduce the hardware overhead significantly because of the reduction of inhibitory connections and refractory control circuits.

In software simulation, the dynamics of excitatory neuron under the control of lateral inhibition can be modeled as the following equation:

$$\tau_V \frac{\mathrm{d}V}{\mathrm{d}t} = (E_{\mathrm{rest}} - V) + \mathrm{int}(g_i == 0)(\lambda g_e) \tag{1}$$

$\mathrm{int}(g_i == 0)$ is equal to '1' if $g_i == 0$ and is equal to '0' if $g_i == 1$. That means only if the inhibition signal is '0' or in low potential, the membrane potential of excitatory neuron can integrate the input spike current. The g_i will be '1' or in high potential during the whole refractory period.

2.2 Memristor-based homeostasis method

The widely used method to achieve homeostasis is based on membrane threshold adaptation. MTA-based homeostasis methods increase the threshold of the excitatory neuron if it spikes too frequently and decrease the threshold if it spikes too rarely. However, it may be hard and costly to realize one adaptive voltage threshold for every neuron, and no hardware solution has been provided for MTA to our best knowledge.

Nowadays, various memristors, including organic devices [31], metal-oxide devices [32], as well as the so-called diffuse memristors and memristors based on metal-insulator transition [33, 34], have been explored for the hardware implementation of artificial synapses. With gradually adaptive resistance, only one memristor device is needed to implement an artificial synapse with adaptive weight, reducing the hardware overhead and improving the energy efficiency [35, 36]. Utilizing the intrinsic adaptive

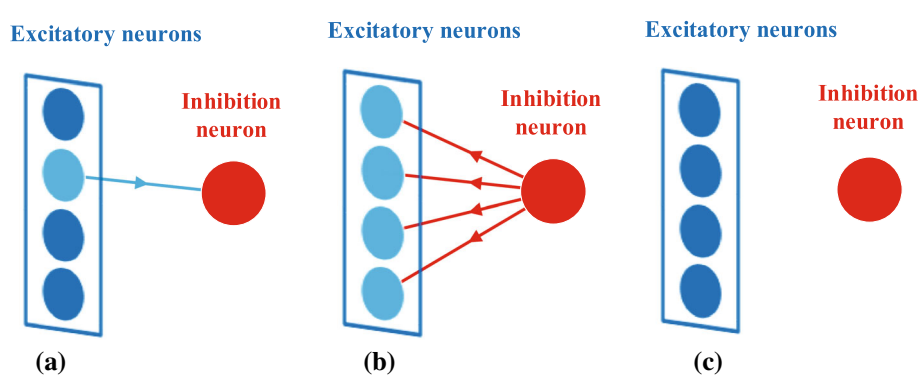

Fig. 1 The schematics of the proposed lateral inhibition method. The connections are only plotted explicitly when transmitting spikes. **a** One learning neuron is excited and sends an excitatory spike to the inhibitory neuron. **b** The inhibitory neuron sends inhibitory spikes to force all the learning neurons turn into refractory state. **c** All the learning neurons return to the normal state after the same refractory period

conductance of memristor, we propose a novel memristor-based homeostasis method as shown in Fig. 2. The proposed method adds an artificial neuron to adapt the leaky current of the excitatory neurons in one layer. The artificial neuron is named as homeostasis neuron in this paper.

The homeostasis neuron is connected to all the excitatory neuron in the same layer via memristors, and its membrane potential is always kept at the resting level E_{rest}. As the membrane potential of the excitatory neuron is larger than the resting level, there will be a leaky current flowing from the excitatory neuron to the homeostasis neuron through the memristor. By changing the conductance of the memristor, the amplitude of the leaky current can be adapted to achieve the homeostasis. When there is no spike between the excitatory neuron and homeostasis neuron, the voltage on the memristor is not high enough to change its conductance. If the excitatory neuron is excited, it will generate a spike to the homeostasis neuron, and the amplitude of the spike voltage is high enough to increase the conductance of the connected memristor. In addition, the homeostasis neuron is designed to generate spikes at a constant frequency (10 Hz in our simulation). The spike voltage pulse generated by the homeostasis neuron is low enough to decrease the conductance value of the memristors. The decrease value of the conductance induced by the spike from homeostasis neuron is much smaller than the increase value induced by the spike from excitatory neurons.

The spike from the homeostasis neuron can keep the memristor conductance at a normal level if the excitatory neuron does not spike too frequently. However, if the excitatory neuron spikes too frequently, the leaky conductance will increase a lot, then it will be difficult for the neuron to cross the threshold, and its spiking rate will decline. Reversely, those neurons that have rarely fired will have lower leaky conductance and respond to the input patterns with larger opportunity. As a result, homeostasis is achieved among the neurons in the same layer. Precisely, this method can be modeled as the following equations, and the conductance change model is inspired from the memristor measurement [37, 38].

$$\tau_V \frac{\mathrm{d}V}{\mathrm{d}t} = g_l * (E_{\text{rest}} - V) + int(g_i = 0)(\lambda g_e) \tag{2}$$

$$\begin{aligned} \Delta w_+ &= \alpha_p \exp\left(-\beta_p \frac{w - w_{\min}}{w_{\max} - w_{\min}}\right) \\ \Delta w_- &= \alpha_d exp\left(-\beta_d \frac{w_{\max} - w}{w_{\max} - w_{\min}}\right) \end{aligned} \tag{3}$$

g_l is the conductance of the memristor between the homeostasis neuron and excitatory neuron. g_l decreases by the value Δw_- after every spike from the homeostasis neuron and increases by the value Δw_+ after every spike from the connected excitatory neuron. The memristor model is as the same as the one for STDP with different parameters of $\alpha_p(1\text{e}{-2})$, $\alpha_d(4\text{e}{-4})$, $w_{\max}(5.0)$, and $w_{\min}(1.0)$ [15].

2.3 Hardware implementation discussion

In this subsection, we discuss the hardware implementation of our proposed methods of lateral inhibition and homeostasis compared with the classical ones. Figure 3 shows the network structure of three lateral inhibition methods proposed in [15, 19] and this work, respectively. Assuming N is the number of excitatory neurons in one layer, the connection number induced by the three different lateral inhibition methods are $N * (N - 1) * 2$, N^2,and $2 * N$, respectively. When it comes to a learning layer with 100 excitatory neuron, the connection numbers of the methods proposed by Querlioz et al. and Diehl et al. increase explosively to 19,800 and 10,000, respectively. While, our proposed method increase linearly to only 200. In addition, our proposed method is implemented on the basis of only one refractory control circuit which is shared by all the excitatory neurons in one layer. Therefore, compared with the lateral inhibition method presented in [15, 19], our proposed method can reduce the hardware overhead

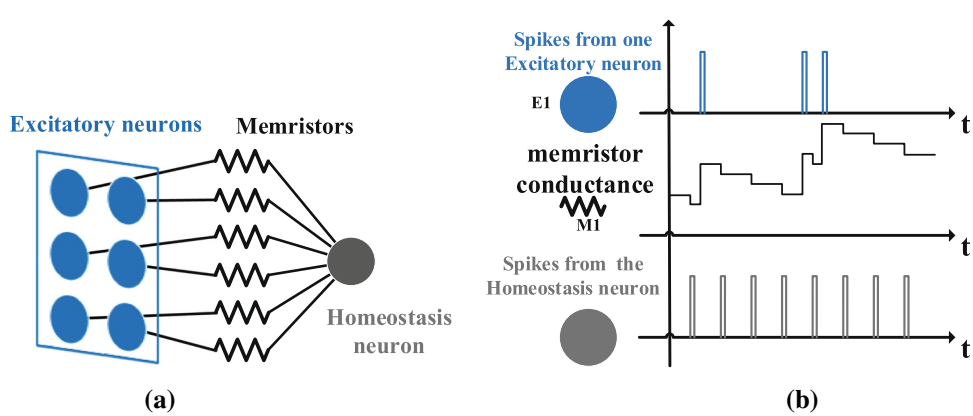

Fig. 2 The schematic of our proposed homeostasis method. **a** The network structure for implementing the proposed homeostasis method. **b** The conductance change of memristor programmed by spike voltage from one excitatory neuron and the homeostasis neuron

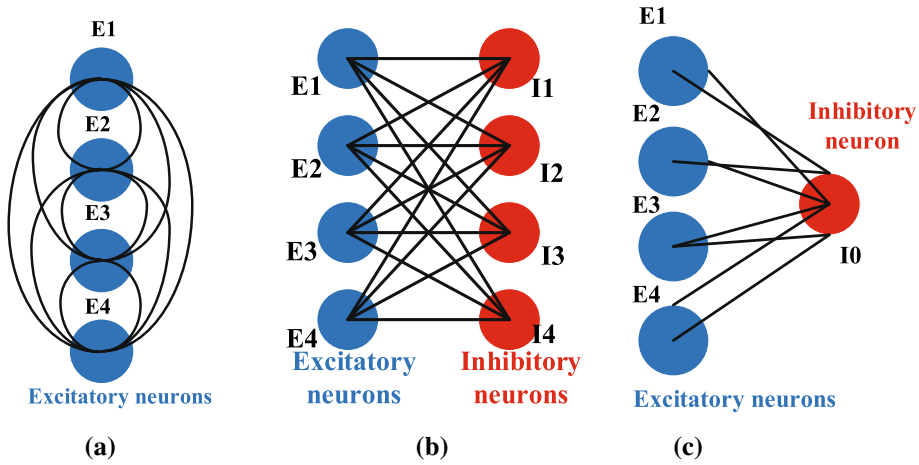

Fig. 3 The network structures of three lateral inhibition methods. The direction of the connections is not plotted. **a–c** Proposed in [15, 19] and this work, respectively

significantly because of the reduction of connections and the sharing of refractory control circuits.

As for the hardware implementation of homeostasis, there has been no hardware solution provided to implement the method based on MTA. Bartolozzi et al. [39] proposed the circuits to implement the homeostasis method based the global synaptic plasticity. The synaptic plasticity is adapted according to the average spiking frequency, which is same with the method in [15], and the adaption circuits for every neuron is so complicated with nearly thirty CMOS devices, while our proposed method for homeostasis can be implemented using only one memristor per learning neuron.

The main drawback of our proposed methods is that they are not exactly biological plausible. However, considering that the powerful and energy-efficient artificial intelligence is our final goal, the bio-inspired learning mechanism should be deployed in an hardware-friendly manner for realistic applications.

3 Fundamental models of SNN-STDP

Here, we describe the models of neuron and synapse with the learning rule of STDP used in the software simulation. Most of the neural models are the same with the models used in [15], except the model of synapse weight when transmitting spikes. Different competitive learning methods are all evaluated based on the same neural models described below for fair comparison.

Neuron model In software simulations, the spiking neuron is typically abstracted as a variable of membrane potential [40]. There are two classes of spiking neurons in terms of the spike effect on post-synaptic neurons. Excitatory spiking neurons send spikes that can increase the membrane potential of post-synaptic neurons, and inhibitory spiking neurons send spikes that can decrease the membrane potential of post-synaptic neurons. When the membrane potential crosses the threshold, a post-synaptic spike will be generated and transmitted to post-synaptic neurons via synapses. After firing, the neuron will come into a refractory period and its membrane potential will be reset to the resting level. During the refractory period, the membrane potential will not be affected by the coming spike and no spike will be generated by the neuron until the end of the refractory period.

The leaky integrate-and-fire(LIF) model is a widely used model that are both biologically plausible and computationally efficient [15, 24]. Before entering the refractory period, the membrane potential of an LIF neuron can be modeled using the following differential equation:

$$\tau_V \frac{\mathrm{d}V}{\mathrm{d}t} = (E_{\mathrm{rest}} - V) + \lambda g_e \tag{4}$$

V is the variable of membrane potential and τ_V is the time constant. E_{rest} is the resting level of membrane potential, and the leaky item $(E_{\mathrm{rest}} - V)$ in the differential equation forces the membrane potential return back to the resting level. g_e is the sum of weights of all the connected excitatory synapses that are transmitting spikes. λ is a multiplicative constant for the synapse weight. The membrane potential is reset to the resting level after firing and stays unchanged during the refractory period.

In our software simulations, the inhibitory spiking neuron is not modeled as the LIF model. The inhibition operation is performed using explicit control spikes rather than inhibitory analog spikes. In addition, only one spike from the excitatory neuron can activate the inhibitory neuron, and the activated inhibitory neuron will send inhibition spikes to all the connected excitatory neurons. When the excitatory neuron receives the inhibition signal, its membrane potential will be reset to the resting level and the neuron will turn into the refractory period [15].

Synapse model The synapse is the connection between two neurons for the transmission of spikes [41]. Typically, the synapse is modeled as a variable conductance that can generate current when applied spike voltage pulses and change its value under the rule of STDP [15, 19]. The value of conductance represent the synapse weight, which contains the information of the learned pattern. When transmitting spikes, the synapse weight can be simulated by solving the following equation:

$$\tau_g \frac{\mathrm{d}g}{\mathrm{d}t} = -g + \delta(s)w \tag{5}$$

g represents the conductance or the weight of the synapse, τ_g is the time constant, $\delta(s)$ is the Dirac delta function, s is the time of spikes, and w is the weight of the synapse. When there comes a spike, the conductance of the synapse will be set to the value of the weight, and then the conductance will decay exponentially to zero to mimic the physical behavior of spikes in biological synapses, while the synapse weight model in Ref. [15] is formulated as the rectangular pulse.

STDP Inspired by biological neural systems, spike-timing-dependent plasticity is the most popular unsupervised learning mechanism for SNNs [42–44]. It changes the weight of synapse according to the time difference between pre- and post-synaptic spikes [45]. Querlioz et al. first proposed a biological-plausible synapse model based on the physical characteristic of memristor [46, 47].

Memristor is a kind of nonvolatile memory device, and its conductance can be changed continuously by voltage pulses with specific amplitude. Being nonvolatile and adaptive, memristor has become the most promising nano-device for the hardware implementation of synapses [48]. Through applying voltage pulses with specific amplitude, the conductance of memristor can be changed as the following equation which is similar to the rule of STDP.

$$\Delta w = \begin{cases} \alpha_p exp\left(-\beta_p \dfrac{w - w_{\min}}{w_{\max} - w_{\min}}\right) & t_{\mathrm{pre}} <= t_{\mathrm{post}} \\ \alpha_d \exp\left(-\beta_d \dfrac{w_{\max} - w}{w_{\max} - w_{\min}}\right) & t_{\mathrm{pre}} > t_{\mathrm{post}} \end{cases} \tag{6}$$

Δw is the modification value of the conductance. α_p, α_d, β_p and β_d are the parameters to model the physical character of the memristor. w is the weight value, and $w_{\max}$ and $w_{\min}$ are the maximum and minimum values of the weight, respectively. t_{pre} and t_{post} are the spike times of the pre- and post-synaptic neuron, respectively.

For the simulation efficiency, STDP is typically triggered by the post-synaptic spike rather than by both the pre- and post-synaptic spikes. The simulation results in [19] showed that only performing the STDP on the post-synaptic spike can achieve comparable performances with performing the STDP on both pre- and post-synaptic spikes. For the simplicity of software simulation and hardware implementation, the specific spiking time of pre-synaptic spike is modeled as a variable of the synaptic trace [49]. Every time a pre-synaptic spike happens, the synaptic trace is added by 1; otherwise, it decays exponentially as the following equation:

$$\tau_{\mathrm{pre}} \frac{\mathrm{dpre}}{\mathrm{d}t} = -\mathrm{pre} + \delta(s) \tag{7}$$

pre is the synaptic trace for the pre-synaptic synapse, τ_{pre} is the time constant, $\delta(s)$ is the Dirac delta function, and s is the time of pre-synaptic spikes. Then, the time difference can be estimated by reading the synaptic trace value when post-synaptic spike happens.

All parameters of the neural models are listed in Table 1.

4 Simulation results and discussions

Firstly, we present the functional evaluation of our proposed competitive learning methods. Secondly, observing that homeostasis method can effect the learning efficiency, we evaluate the learning efficiency of three different homeostasis methods including our proposed memristor-based one and the two MTA-based ones proposed in [15] and [19]. The evaluation of the three homeostasis methods are all based on the same software simulation platform of Brian [50, 51]. Brian is an open-source SNN simulator based on Python and is available on almost all platforms [51]. It can provide an uniform software simulation framework for different learning algorithms with flexible and extensible models. We reproduced the learning algorithm of Querlioz's work in Brian, and the reproduced algorithm in Brian can achieve the same performances as reported in [15]. Another two complete learning algorithms using Diehl's [19] and our proposed homeostasis methods are established based on the same neural models introduced in Sect. 3.

The widely studied case of MNIST dataset is used as the benchmark task [52]. The MNIST dataset contains 60000

Table 1 Parameters of the network model

E_{rest}	0 V	V_{th}	0.5 V
τ_V	100 ms	τ_g	1 ms
α_p	8e−3	α_d	6e−3
β_p	3	β_d	3
Maximum weight	1.0	Minimum weight	0.0
τ_{pre}	25 ms	λ	1.0

training and 10,000 testing grayscale handwritten number images belonging to ten categories of '0–9'. The input is coded by rate coding scheme where the absolute value of the pixel is converted to the spike rate of the Poisson spike generator in Brian [51, 53]. The maximum spiking rate, corresponding to the highest grayscale value of 255, is 63.75 Hz. The time window for every input examples is 500 ms, including 350 ms presenting the input spikes and 150 ms without any spikes. The no-spike input window can be deleted with a little performance degradation. No pre-processing of the MNIST dataset is needed.

4.1 Functionality evaluation of the proposed methods

Figure 4 shows the network module for competitive learning combining our proposed methods of lateral inhibition and homeostasis. Except for the excitatory neurons, one inhibition neuron and one homeostasis neuron are added to achieve the lateral inhibition and homeostasis, respectively. The inhibitory neuron is bi-directionally connected to all the excitatory neurons to perform lateral inhibition. The homeostasis neuron is connected to all the excitatory neuron via memristors to achieve homeostasis through adapting the conductance of the memristor. It should be noted that our proposed network module can be used as a learning layer in full connected architecture or a neuronal map of convolutional kernel in convolutional architecture [15, 24].

In this paper, we instantiate our proposed network module as a fully connected single-layer network for functional evaluation. The input is fed directly to learning layer populated with excitatory neurons. Every pixel of the input image is represented by one Poisson spike generation neuron whose spike rate is proportional to the absolute pixel value. The learning layer is populated with excitatory LIF neurons and is fully connected by weighted connections to the input spike generation neurons. The range of the weight value is [0.0, 1.0], and the initial weights before learning are assigned around the normal distribution with the media value of 0.5. Input patterns are learned through adapting the synapse weights between the input layer and learning layer under the rule of STDP.

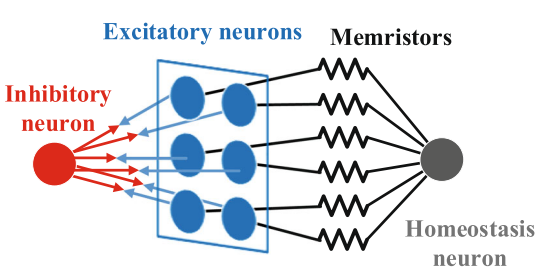

Fig. 4 The network architecture of our proposed network module for competitive learning. The number of excitatory neurons is variable depending on the application

After the training procedure, every excitatory neuron is scored for every class according to its spike number as the following equation:

$$\mathrm{Score}_{N_c} = \frac{\sum \mathrm{spike}_{N_c}}{\sum \mathrm{spike}_N} \tag{8}$$

Score_{N_c} and spike_{N_c} represent the score and spike number of the Nth neuron for the class of c, respectively. spike_N represents the spike number of the Nth neuron for all the classes during the training procedure. The score is calculated at the end of the training procedure using only the last 1000 training examples. We do not label the excitatory neuron to only one specific class, because we believe that the excitatory neuron learns general patterns rather than a specific digit. During the test procedure, STDP learning is stopped and the scored neurons are then used to predict the input example. The label of the input example is predicted based on the highest sum of scores of which class the neurons have fired for. The classification accuracy is the percentage of the number of correctly predicted examples and the total number of test examples.

Firstly, a network of 10 excitatory neurons is simulated to validate the functionality of the proposed competitive learning methods. Figure 5 visualizes the weight learned by the ten neurons trained by the full training dataset. It can be seen that without any supervision the ten neurons learned distinct patterns from the digits. The classification accuracy of the ten neuron network converges to value of 62%.

To validate the effect of lateral inhibition, Fig. 6 presents the weight visualization of the network trained without the lateral inhibition method. It can be seen that some neurons learned nearly the same patterns, which decrease the learning capacity of the whole network.

To show the effect of homeostasis, we simulated two network of 10 and 50 learning neurons with or without homeostasis implemented. Figure 7 presents the standard deviation of spike number of different learning neuron and the testing accuracy during the training procedure of 60,000 training examples. The standard deviation of the spike number is calculated using the following equation:

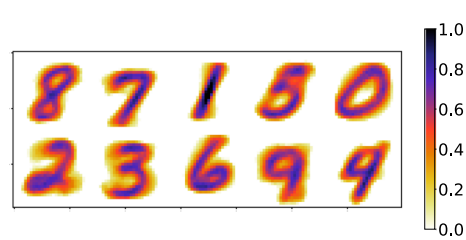

Fig. 5 Weight visualization of the ten neurons trained with one iteration of the full training dataset

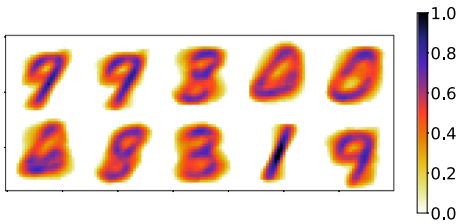

Fig. 6 Weight visualization of the ten neurons trained without lateral inhibition using one iteration of the full training dataset

$$\mathrm{SD}_{\mathrm{spike}} = \sqrt{\frac{\sum_{i=1}^{N}(\mathrm{spike}_i - \mathrm{spike}_{\mathrm{average}})^2}{N}} \tag{9}$$

$\mathrm{SD}_{\mathrm{spike}}$ is the standard deviation of spike numbers of different neurons, spike_i is the spike number of one neuron, $\mathrm{spike}_{\mathrm{average}}$ is the average spike number of all the neurons during the training procedure. The 60,000 training examples are divided into 30 training iteration of 2000 examples. Testing accuracy is evaluated on the full testing dataset of 10,000 examples after every training iteration. It can be seen that the spike number deviation under the control of the homeostasis method is much smaller than that without homeostasis method. The learning speed and accuracy of the network with homeostasis method are better than those without homeostasis method. Therefore, the proposed homeostasis method is effective to mitigate the inhomogeneity. It can also be seen that homeostasis method can play an important role in effecting the performance of the SNN-STDP learning algorithm.

4.2 Performance evaluation

We observe that the method to achieve homeostasis can play an important role in affecting the learning efficiency. If the method to achieve homeostasis does not mitigate the inhomogeneity timely and moderately, some well-trained neurons will always inhibit the undertrained neurons from learning, thus reducing the learning efficiency significantly. However, the effect of the homeostasis method on learning efficiency has never been studied [54, 55]. In this subsection, simulation results comparing our proposed homeostasis method with two classical MTA-based ones are presented considering the learning efficiency and classification accuracy. Three learning algorithms are established based on the same neural models, including the model of neurons, synapses, and STDP. The only difference among the three learning algorithms are the methods to achieve homeostasis.

Firstly, we test the performances of the three different learning algorithms with various numbers of excitatory neurons less than 100 neurons. To evaluate the learning efficiency, the performance is tested on the test dataset after every 2000 training examples presented. As plotted in Fig. 8, the learning algorithm with our proposed homeostasis method can learn faster than the method proposed by Querlioz and achieve 6–10% higher performance than the method proposed by Diehl. According to the examples needed to achieve 80% and 85% accuracy for 50- and 90-neuron network, our proposed method can learn more than 30–50% faster than Querlioz's method. 80% and 85% are the typical accuracy for 50- and 90-neuron SNNs, respectively.

To find the reason for the differences on learning efficiency and performance, we present the standard deviation of spike number of the excitatory neurons in Fig. 9.

It can be seen that, corresponding to the high learning efficiency, both our proposed method and Diehl's method achieve smaller spike number deviations compared with Querlioz's method. Small spike number deviation indicates that every neuron has nearly the same opportunity to learn

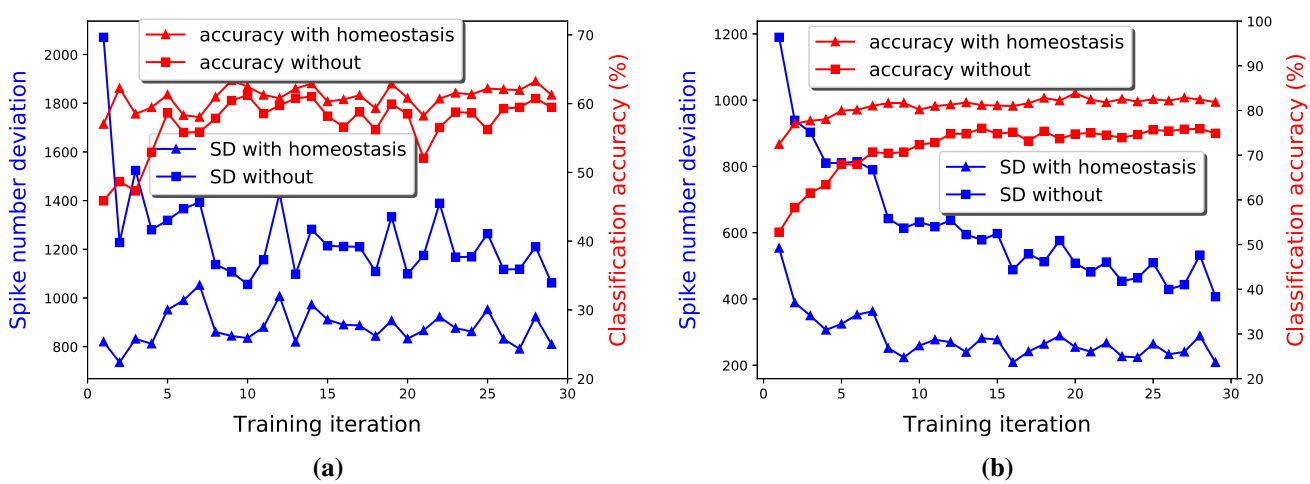

Fig. 7 Standard deviation of spike number and testing accuracy of network with 10 learning neurons (**a**) and 50 learning neurons (**b**) with or without the homeostasis method

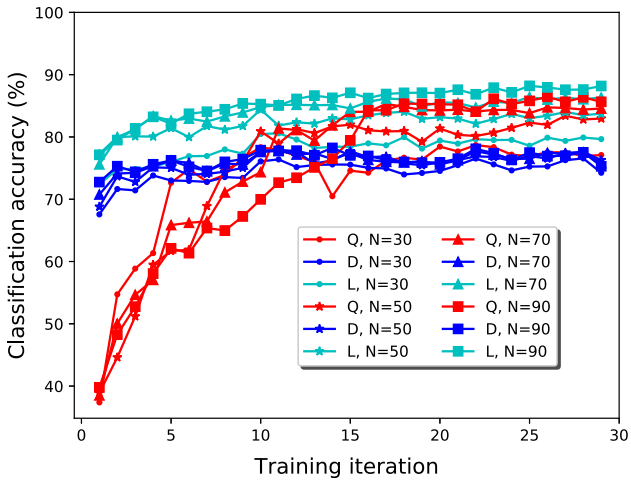

Fig. 8 The performance after every iteration of 2000 training examples. The performances achieved by Querlioz's, Diehl's and our proposed methods are labeled using '*Q*,' '*D*' and '*L*,' respectively. *N* indicates the number of excitatory neurons

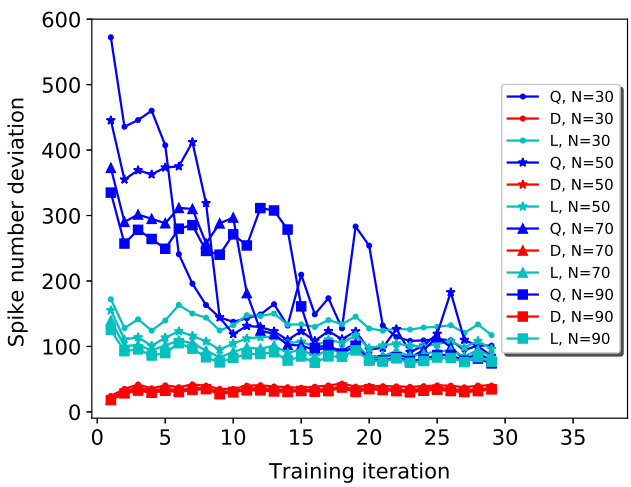

Fig. 9 The spike number deviation during the learning procedure. The data for Querlioz's, Diehl's and our proposed methods are labeled using '*Q*,' '*D*' and '*L*,' respectively. *N* indicates the number of excitatory neurons

during the training procedure. Therefore, less extra examples are wasted on the well-trained neurons, and the undertrained neurons can get enough examples to learn. This help to improve the learning efficiency by reducing the number of training examples required and improve the performance by balancing the learning progress of different neurons. At the beginning, the spike number deviation of Querlioz method is much larger than our proposed method and Diehl's method. It may because that both our proposed method and Diehl's method perform the adaptation according to the real time spike, while Querlioz's method performs the adaptation according to the average spiking frequency, which is not stable at the beginning.

The reason for the low performance of Diehl's method is that the adapted threshold is too high for some pattern to activate the learning neuron. In both our simulations and Diehl's origin experiments, the adapted threshold increased from 0.5 V to a value around 1.0 V. As a result, the input numbers with less spikes, like '1' and '7,' have smaller opportunity to activate the learning neuron, while the input numbers with more spikes, like '0' and '8,' have larger opportunity to activate the learning neuron. The weight matrix trained using Diehl's methods with 90 learning neurons is plotted in Fig. 10. Of the 90 learning neurons, 17 neurons learned the pattern of '0,' 12 neurons learned the pattern of '8,' while only 4 neurons learned the pattern of '1,' 5 neurons learned the pattern of '7.' It can be seen that the adapted threshold induces new inhomogeneity among different patterns. Therefore, the change amplitude of the adaptive variable should be limited. The change amplitude of the conductance in our proposed method is naturally limited because of the intrinsic physical character of memristor. Thus, our proposed method can achieve higher performance than Diehl's method.

To compare with the state-of-the-art performance on the task of digital recognition achieved by single-layer network, we tested the performances of our proposed method with larger number of learning neurons of 100 and 300. As for a network with 100 learning neurons, our proposed methods can achieve a classification accuracy of larger than 88% with less than two iteration of the training dataset, while the network proposed by Diehl can only achieve a classification accuracy of 82%, and Querlioz's proposal needed three iteration of the training dataset to train the network to achieve a classification accuracy of

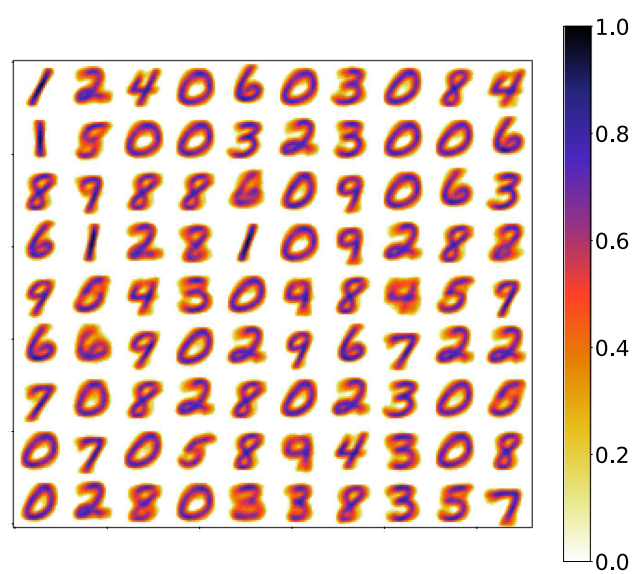

Fig. 10 The weight matrix of 90 neurons trained using the homeostasis method proposed by Diehl after presenting one iteration of the full training dataset

Table 2 Classification accuracy of state-of-the-art SNN-STDP learning algorithms on MNIST dataset

SNN topology	Computation	Input coding	#Training parameter	Accuracy (%)
Fully connected one-layer [15]	Spiking neuron, STDP	Rate coding	235,200	93.5
Fully connected two-layer [19]	Spiking neuron, STDP, weight normalization	Rate coding	5,017,600	95
Fully connected one-layer (this work)	Spiking neuron, STDP	Rate coding	235,200	92
Convolutional three-layer [24]	Spiking neuron, STDP, DoG filter, convolutional, max-pooling, weight sharing	Rate coding	25,488	91.1
Convolutional five-layer [21]	Spiking neuron, STDP, DoG filter, convolutional, max-pooling, weight sharing, classifier	Time coding	75,750	98.4

85%. As for a network with 300 learning neurons, less than two iteration of the full training dataset is needed for training to achieve an test accuracy of 92%.

For presenting broader comparison, we list the performance and properties of state-of-the-art SNN-STDP algorithms on MNIST dataset in Table 2. It should be noted that the mechanisms of lateral inhibition and homeostasis are all used in the listed SNN-STDP algorithms, except that homeostasis is not explicitly announced in [21].

Among the fully connected architecture, Diehl's algorithm achieved the highest performance. However, the accuracy of 95% is achieved using 6400 learning neurons, and weight normalization is used during the training procedure, which is not easy to be implemented on hardware. Compared with fully connected architecture, convolutional architecture can achieve better accuracy with much less training parameters. It is because that the global feature in convolutional architecture is extracted hierarchically, and the weight kernel size is much smaller than that of fully connected architecture. Therefore, convolutional architecture is more efficient to arrange learning neurons. However, convolutional SNN-STDP introduces some components that are not biological plausible from the view of neuroscience. On the other hand, these DNN-like components are hard or costly to be implemented on neuromorphic hardware, such as weight sharing and convolutional operation. In conclusion, convolutional architecture is more efficient and powerful than fully connected architecture, especially for complex tasks, while the existing convolutional SNN-STDP algorithms are not the final solution for SNN-based artificial intelligence system. The main concern of this paper is to provide efficient implementation method of lateral inhibition and homeostasis, which can be used for different network architectures. Considering that lateral inhibition and homeostasis are inspired from biological neural systems, our proposed methods are expected to be used in more futuristic SNN-based algorithms.

5 Conclusion

Lateral inhibition and homeostasis are two essential mechanisms to achieve competitive learning for SNNs trained by STDP. However, the existing methods of lateral inhibition and homeostasis are not efficient to be implemented on neuromorphic hardware. In this paper, targeting hardware-friendly solutions, we proposed two novel methods of lateral inhibition and homeostasis with hardware in mind. The proposed lateral inhibition method can reduce the hardware overhead greatly by reducing connection number from N^2 to $2N$ and sharing refractory control circuits among neurons in the same layer. Based on memristor, we proposed a novel homeostasis method by adapting the leaky current. Compared with previous homeostasis methods, with no hardware solution provided, our proposed method can be implemented easily using one memristor device for every excitatory neuron. For the first time, we studied the effect of homeostasis method on learning efficiency by simulating three SNN-STDP learning algorithms with different homeostasis methods. Software simulations on the recognition task of MNIST dataset are carried out to evaluate the performance of the SNN-STDP learning algorithms. Simulation results show that, compared with two classical homeostasis methods, our proposed method can achieve higher learning efficiency while maintaining comparable performances.

Acknowledgements This research is funded by the National Key R&D Program of China [Grant No. 2018YFB2202603] and in part by the National Natural Science Foundation of China [Grant Nos. 61802427 and 61832018].

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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